



Apache Spark on Amazon EMR

Fast, Distributed Data Processing

AWS Machine Learning Associate • Part 7

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What We'll Cover



What is Apache Spark?



Why Spark is faster than MapReduce



Spark's architecture



Why run Spark on Amazon EMR?



Key features & benefits



Common use cases

What is Apache Spark?



Open-source & distributed

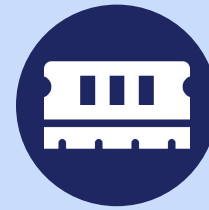
- **Spreads data & computation across a cluster**
Large jobs split into parallel pieces
- **General purpose**
Batch, streaming, SQL, ML (MLlib) & graph processing, in one engine
- **Multi-language**
Python (PySpark), Scala, Java, R, or SQL

Why Spark is Faster



Traditional MapReduce

Writes intermediate results to disk between steps.
Reliable, but slow.



Apache Spark

Keeps data in memory across steps whenever possible,
only touching disk when it has to.

Spark's Architecture



Together, these let Spark scale a single job across dozens or even thousands of machines.

Why Run Spark on Amazon EMR?



**AWS's managed
big data platform**

- **EMR provisions & tunes clusters for you**
No manual setup, install, or configuration
- **Reads & writes straight from Amazon S3**
Storage decoupled from compute — data persists after shutdown
- **Flexible deployment**
EMR on EC2, EMR Serverless, or EMR on EKS

Key Features & Benefits



Auto-scaling

Grows or shrinks the cluster as workload demands change



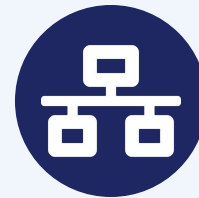
Spot instance support

Cuts costs significantly for interruption-tolerant jobs



EMR Studio & Notebooks

Interactive, Jupyter-based development against your cluster



Deep AWS integration

Glue Data Catalog, Athena, SageMaker & CloudWatch

Common Use Cases



Large-Scale ETL

Cleaning & transforming massive datasets before they land in a warehouse or lake



Streaming Analytics

Processing real-time data as it arrives with Spark Streaming



Machine Learning Pipelines

Feature engineering & training models at scale with MLlib



Interactive Data Exploration

Ad hoc analysis directly through notebooks

Recap

- Spark: a fast, in-memory, general-purpose processing engine.
- EMR: AWS's managed way to run Spark, no cluster babysitting.
- Together: scalable ETL, streaming, and ML, without the ops burden.

Thanks for watching!